

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 24)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 1

1

Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 1

2

Give **three** limitations of kick sampling.

[3 marks]

1

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2

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3

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0 1

3

Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 1

4

Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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10

0 2 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

0 2 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

0 2 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

0 3 . 1

Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

03 . 2

Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

03 . 3

Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

1

2

3

10

04 . 1

Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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04 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

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04 . 3

Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

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Water flow

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Day length

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Temperature

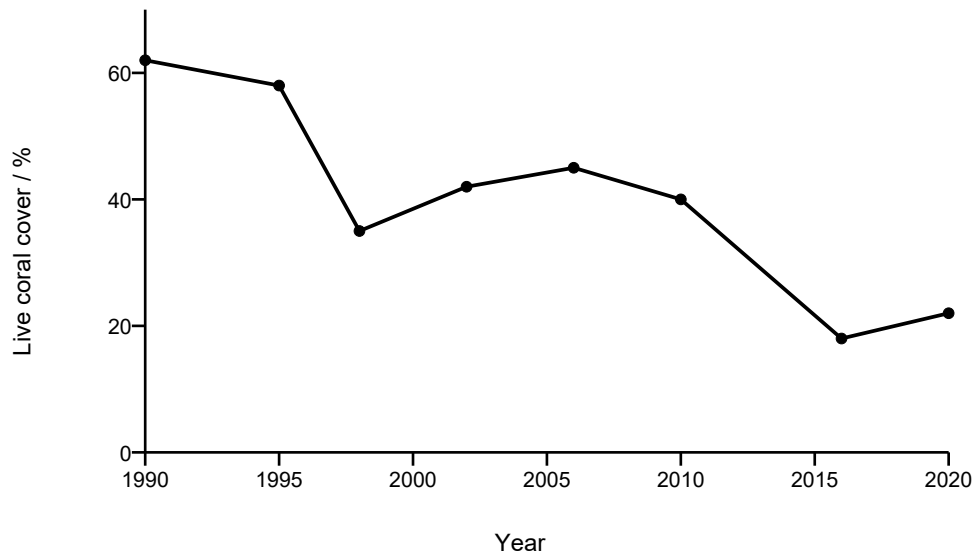
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10

0 5

Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 5

1

Describe the changes in coral cover shown in Figure 1.

[2 marks]

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0 5

2

Explain what coral bleaching is and give **three** causes.

[4 marks]

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0 5 . 3

Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

0 5 . 4

Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

10

0 6 . 1

Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

1

2

3

4

0 6 . 2

Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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06 . 3

Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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10

07 . 1

Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light

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Dissolved oxygen

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Nutrients

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07 . 2

Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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10

08 . 1

Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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08 . 2

Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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08 . 3

Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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09

A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

09

1

Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

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09

2

Explain the changes shown in Table 1.

[5 marks]

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09 . 3

Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

09 . 4

Describe the early stages of a lithosere.

[1 mark]

10

10 . 1

Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

[illegible]

10 . 2

State what is meant by **biodiversity**.

[1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

END OF QUESTIONS